

Chronic Hepatitis — HBV Serology



1. HBsAg suspect — first to appear (gold star)

WHAT YOU SEE

Detective arresting the 'HBsAg' suspect with a gold star; 'first to appear' tag, center.

WHAT IT ENCODES

GOLD STAR: HBsAg (surface antigen) is the FIRST marker to appear and signals ACTIVE infection (acute or chronic). Persistence >6 months = chronic HBV. It is the screening test; its presence means the patient is infectious.

2. HBsAg lineup figures

WHAT YOU SEE

Police-lineup silhouettes labeled HBsAg / anti-HBs / HBsAg / anti-HBc IgG along the back wall, left.

WHAT IT ENCODES

The lineup = the marker panel you assemble to stage HBV: HBsAg (infection), anti-HBs (immunity), anti-HBc IgM vs IgG (acute/window vs past/chronic), HBeAg/anti-HBe (replication). Reading the combination, not any single band, gives the diagnosis.

3. HBV DNA child

WHAT YOU SEE

Small figure holding 'HBV DNA', plus a large HBV DNA jar, lower left/center.

WHAT IT ENCODES

HBV DNA = viral load, the direct measure of replication and the main driver of treatment decisions and infectivity. High DNA tracks with HBeAg-positive active disease; low DNA with the inactive carrier state.

4. Vaccinated portrait

WHAT YOU SEE

Framed photo 'vaccinated' (with gold star card), top of board.

WHAT IT ENCODES

VACCINATED = isolated anti-HBs positive ONLY, with NO anti-HBc (and HBsAg negative). The vaccine is recombinant surface antigen, so it never produces core antibody. Anti-HBs ≥10 mIU/mL = protective.

5. Resolved / past infection portrait

WHAT YOU SEE

'resolved past infection' framed portrait, top center.

WHAT IT ENCODES

RESOLVED past infection = anti-HBs POSITIVE + anti-HBc IgG positive (HBsAg negative). The presence of anti-HBc is what distinguishes natural immunity from vaccination. These patients can reactivate under immunosuppression.

6. Chronic active portrait

WHAT YOU SEE

Red 'chronic active' devil-faced portrait holding Rx, top center-right.

WHAT IT ENCODES

CHRONIC ACTIVE = HBsAg positive >6 months + HBeAg positive + HIGH HBV DNA + elevated ALT. High replication, high infectivity → treat (entecavir/tenofovir). HBeAg-negative chronic active hepatitis exists too (precore mutant) with high DNA but negative HBeAg.

7. Inactive carrier portrait

WHAT YOU SEE

'inactive carrier' framed portrait, top right.

WHAT IT ENCODES

INACTIVE CARRIER = HBsAg positive + anti-HBe positive (HBeAg negative) + LOW HBV DNA + NORMAL ALT. Low replication, monitor rather than treat; still needs HCC surveillance if cirrhotic or high-risk.

8. Window period portrait

WHAT YOU SEE

Shadowy 'window period' silhouette frame, far top right.

WHAT IT ENCODES

WINDOW PERIOD = anti-HBc IgM positive ONLY — HBsAg has cleared but anti-HBs has not yet risen. The core IgM is the only positive marker bridging the gap, so it is the clue to recent acute infection during this seronegative window.

9. Anti-HBc IgM badge

WHAT YOU SEE

Gold badge 'ANTIBODY IgM' on the desk, bottom center.

WHAT IT ENCODES

Anti-HBc IgM = ACUTE infection or window period. It is NOT a marker of chronicity. High-titer IgM in acute hepatitis; can transiently rise in severe flares of chronic disease.

BOARD TRAP

'Anti-HBc IgM means chronic HBV' → wrong; IgM indicates acute infection / window period — chronicity is shown by anti-HBc IgG with persistent HBsAg.

10. Anti-HBc IgG badge

WHAT YOU SEE

Gold badge 'ANTIBODY IgG' on the desk, bottom center.

WHAT IT ENCODES

Anti-HBc IgG = PAST or CHRONIC infection (persists for life). With HBsAg = chronic; with anti-HBs and no HBsAg = resolved. Marks anyone ever naturally infected — flags reactivation risk before immunosuppression.

11. WANTED vaccine poster

WHAT YOU SEE

'WANTED' poster with a syringe and red X, left-center of board.

WHAT IT ENCODES

Vaccination produces anti-HBs alone. The trap is calling an isolated anti-HBs patient 'resolved infection' — without anti-HBc it is vaccine-induced immunity, not recovery from disease.

BOARD TRAP

'Isolated anti-HBs positive = resolved past infection' → wrong; isolated anti-HBs with NO anti-HBc = vaccinated, not natural recovery.

12. Ribavirin DECOY (red X)

WHAT YOU SEE

'DECOY' and 'ribavirin' posters with red Xs, right of board.

WHAT IT ENCODES

Ribavirin is an HCV/RSV drug — it has NO role in HBV. HBV first-line is entecavir or tenofovir (± PEG-interferon). Ribavirin in an HBV question is a wrong-disease distractor.

BOARD TRAP

'Treat HBV with ribavirin' → wrong; ribavirin is for HCV — HBV is treated with entecavir/tenofovir, not ribavirin.
